

Research Briefing | Eurozone and UK

How quickly are indirect effects on core inflation fading?

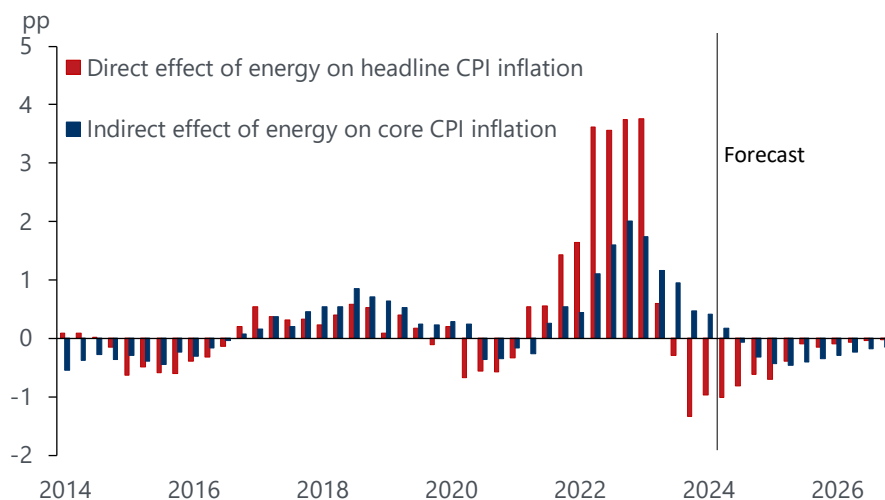
- The large indirect boost to core inflation from the recent surge in energy costs in the eurozone and UK is fading. Marked declines in inflation among energy-intensive core goods and services items have helped reduce core inflation in both economies in recent months.
- The recent sharp drop in wholesale energy costs has already caused the direct effects of energy on headline CPI inflation to turn negative in the UK and eurozone, as household gas and electricity bills fall. However, reflecting the usual lag, the indirect effects from energy prices on core inflation have not yet turned negative. We think this will happen in the next quarter or two.
- The swing in the indirect effects of energy prices – from lifting core inflation recently, to reducing it in the coming quarters – is likely to help pull down core and headline inflation in both the UK and eurozone over the rest of this year and next. We expect headline inflation in both the UK and eurozone will fall below 2% in Q2 or Q3.
- Lower headline inflation, along with greater confidence that inflation expectations are well contained and pay growth is slowing, should allow the European Central Bank and Bank of England to start to cut rates in the next few months.

The direct effects of swings in energy prices on headline inflation, through changes in household costs for electricity, gas, and petrol, added substantially to inflation in the UK and eurozone in 2022 and 2023. The peak effects added about 4ppts to inflation in both economies. These direct effects have recently turned negative as the drop in wholesale energy costs starts to feed through to household electricity and gas bills.

However, as we have [previously noted](#), swings in energy prices also have substantial effects on core inflation (as well as food prices), because energy prices are an important component of firms' costs, especially in the services sector. These indirect effects come through more slowly than the direct effects of energy prices, but – having added to core inflation recently – are likely to put significant downward pressure on core inflation in the coming quarters.

Chart 1: Lower energy prices are likely to dampen core inflation significantly

UK: Effects of energy prices on inflation



Sources: Oxford Economics / Haver Analytics

How quickly are indirect effects on core inflation fading?

Indirect effects of energy prices on UK inflation

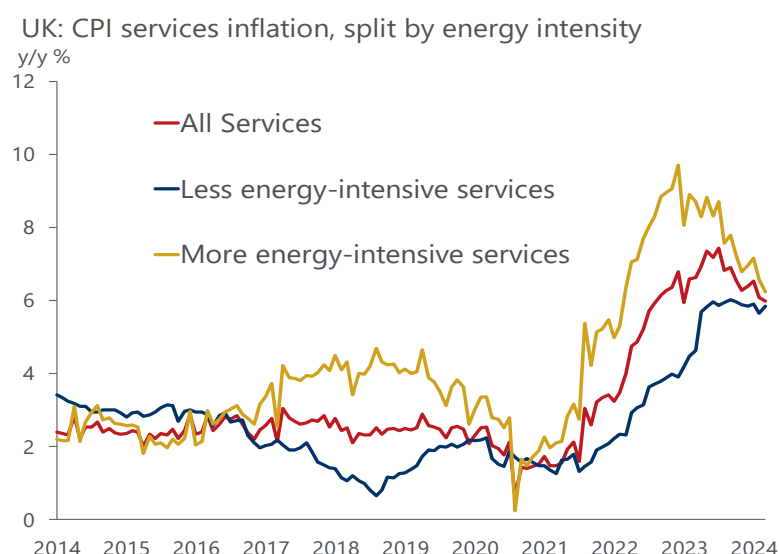
To investigate the effects for the UK, like [our previous work](#), we use a detailed ONS breakdown of the energy intensity of individual CPI items, split into [84 detailed categories](#). These figures suggest that energy costs account for an average of 1.6% of the selling price for core CPI items, although this estimate may slightly understate the total effect because it does not include the indirect energy content of imported goods and services. The 1.6% figure is, of course, well below the direct weight of energy costs in the CPI of about 4.4% (note this excludes the share of household energy costs that is accounted for by taxes and labour costs). But it's still large enough for energy to significantly affect core inflation.

Energy intensity among core CPI items ranges widely. For example, energy intensity is relatively low (less than 0.5% of the price) for insurance, clothing, and footwear, but relatively high (more than 3% of the price) for garden plants, home repair products, air travel, and hotels.

Within core inflation, average energy intensity is markedly higher for services (2.0%) than for goods (1.2%). Moreover, the CPI items with very high energy intensity are overwhelmingly services rather than goods: items with energy intensity of 2.5% or more account for 44% of all CPI services but only 6% of core goods.

Chart 2 shows the weighted average inflation rates among service sector items, split by energy intensity. We measure the indirect effect of energy prices on services inflation as the gap between overall services inflation and the rate excluding items with high energy intensity (i.e., the gap between the red and blue lines). For services, this indirect effect peaked at about 2.5ppts in late 2022, and fell to about 0.4ppts in Q1 this year.

Chart 2: Inflation has slowed sharply for energy-intensive services



Source: Oxford Economics

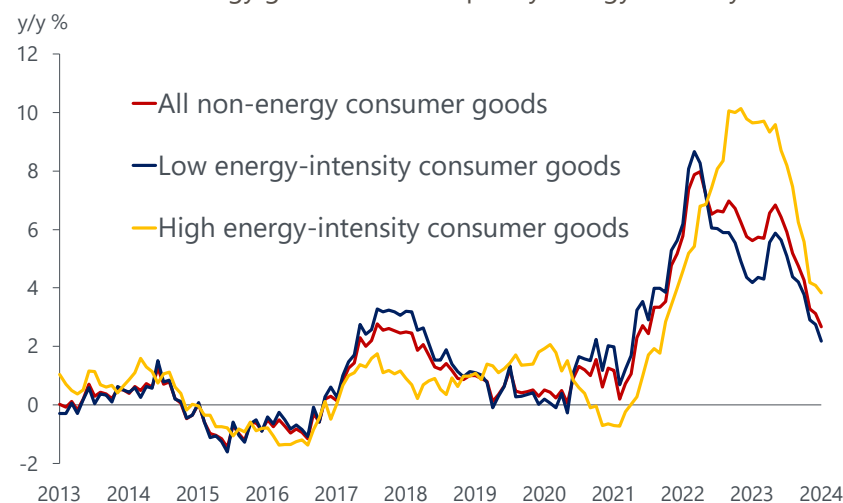
An important inference from this is that it is wrong to regard services inflation as purely reflecting domestic cost and capacity pressures; in fact, it is significantly affected by energy prices, and some parts of services inflation also have been affected by supply shocks to food and goods prices.

We ran a similar exercise for core goods (**Chart 3**). Inflation has slowed markedly among a wide range of core goods items, but the recent drop in inflation has been especially marked among items with relatively high energy intensity. As for services, the indirect effect of energy prices on core goods inflation has diminished, but has not yet turned negative.

How quickly are indirect effects on core inflation fading?

Chart 3: Among core goods, inflation has slowed especially sharply for energy-intensive items

UK: CPI non-energy goods inflation split by energy intensity

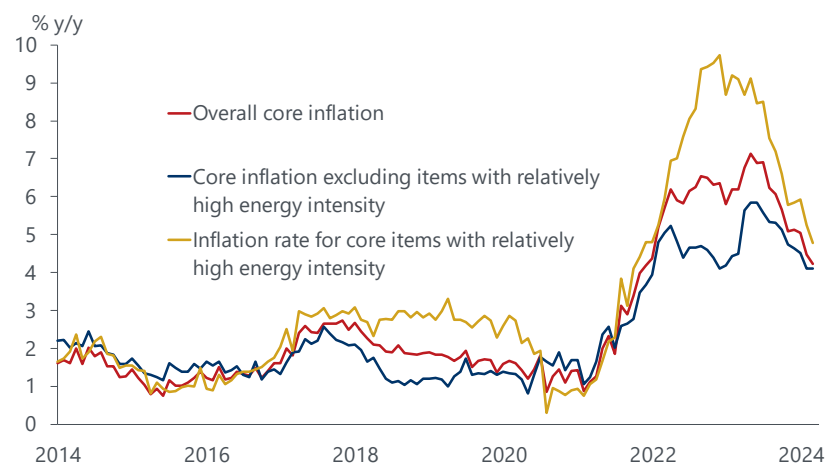


Source: Oxford Economics

We weighted the results for core goods and services by the respective shares of services and goods in the core CPI to get estimates for core inflation split by energy intensity (**Chart 4**). Again, we take the gap between overall core inflation and the rate excluding items with high energy intensity as a measure of the indirect effect of energy prices on core inflation (**Chart 1**). Note that for both goods and services, there was little difference in the average inflation rates for items with high and low energy-intensity in the pre-pandemic period. Hence, the recent wide gap between the two series probably reflects the energy price surge rather than composition effects.

Chart 4: Inflation rose especially sharply among items with high energy intensity

UK: Core CPI inflation split by energy intensity



Sources: Oxford Economics/Haver Analytics

Based on this, we estimate that indirect effects of energy prices added about 2ppts to core inflation in Q4 2022, but this effect fell to about 0.4ppts in Q1 this year. The indirect effects of energy on headline inflation were similar because the energy intensity of food (which is not in core inflation) is relatively high.

There are inevitable uncertainties with our approach, and it may slightly understate the total impact of energy costs on inflation. This is because our methodology assumes that energy prices have no effect on inflation for items with relatively low energy-intensity, whereas they likely have some effect.

Compared to energy's direct effects on inflation, the indirect effects of energy on core inflation come through more slowly: if energy prices rise, the peak indirect effect is about three quarters after the peak

How quickly are indirect effects on core inflation fading?

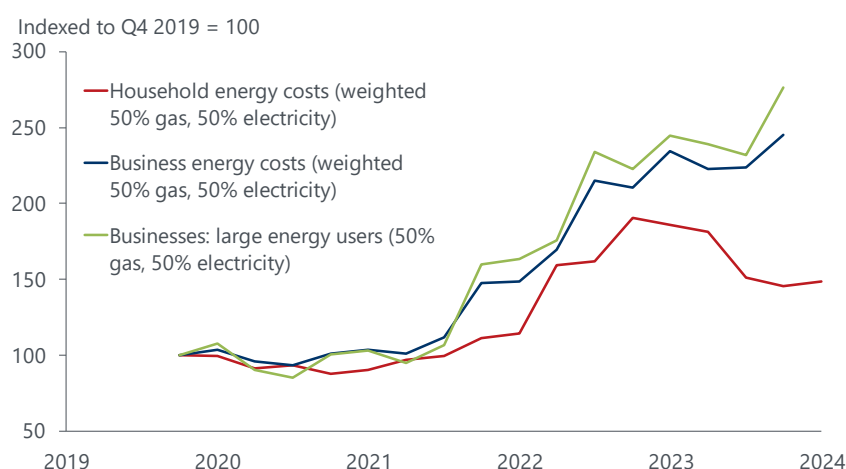
direct effect, and fades gradually over a couple of years. This is partly because some firms have longer-term fixed-price contracts, whereas most household energy prices reset every quarter. It's also because of the lag in the pass-through from changes in firms' energy costs to their selling prices.

Moreover, the relation between the direct and indirect effects of energy prices varies over time because energy prices for firms and households don't always move in lockstep. In particular, the household energy price caps that were implemented by the UK government over 2022-2023 did not apply to firms.

The different dynamics of direct and indirect pass-through from energy prices have been important recently because energy costs rose more for firms than households in 2022-2023 and have been slower to fall. For example, in Q4 2023, household electricity prices were up by 61% and gas prices up by 71% from their Q4 2019 levels. However, the average increases for firms were 103% for electricity and 187% for gas – among heavy energy users, the increases were 130% for electricity and 222% for gas.

Chart 5: Energy costs have risen by more for businesses than households since 2019

UK: Change in energy costs for households and businesses



Sources: Oxford Economics/Haver Analytics

As a result, the contrast between the direct and indirect effects of energy prices on inflation is marked at present. Wholesale gas and electricity prices are down about 75% from the 2022 average. Some of this has already come through to household gas and electricity charges. As a result, the direct effect of energy prices on headline inflation has fallen from +3.7ppts in Q4 2022 to -1.0ppt in Q1 this year, a swing of nearly 5ppts. On our estimate, energy's indirect effect on core inflation has fallen from 2.0ppts to 0.4ppts over the same period, but remains positive. So the indirect boost to inflation from rising energy costs is fading, but the downward pressure on inflation from recent declines in wholesale energy prices has not yet fed through.

Assuming energy prices follow the path implied by forward markets, the direct effect on headline inflation from energy will remain significantly negative through this year, and revert to around zero during 2025. As the recent sharp decline in wholesale gas prices and electricity prices feeds through to firms' energy costs – and firms then largely pass these lower costs onto their selling prices – the indirect effects of energy on core inflation are also likely to turn negative during this year and cut nearly 1ppt off core inflation over the next year. In other words, the average inflation rate for energy-intensive items is likely to fall well below that for items with relatively low energy-intensity, pulling down overall core inflation.

Given the lags between changes in wholesale prices and firms' energy costs, as well as lags between firms' costs and the effect on their selling prices, the indirect effects of energy on inflation probably will remain significantly negative through next year.

These downward effects on headline and core inflation from energy prices are likely to be reinforced by continued weakness in consumer goods prices (as supply chain effects unwind), as well as disinflationary impacts from spare capacity and slowing pay growth. In all, we expect CPI inflation will return to the 2% target in April or May and probably will be a little below 2% on average in the second half of this year.

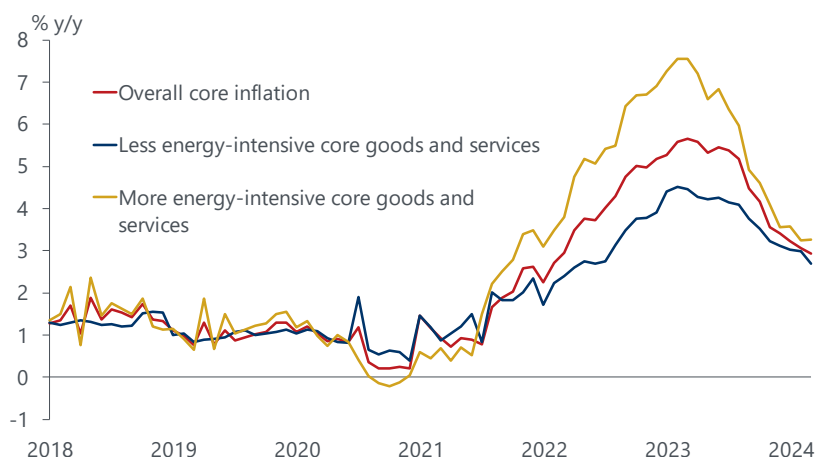
How quickly are indirect effects on core inflation fading?

Indirect effects of energy prices on eurozone inflation

It's a similar story for the eurozone, albeit with a lower trend for services items with relatively low energy intensity. Using [this ECB decomposition](#) of CPI items by energy intensity, we estimate that the indirect effect of energy prices on core inflation peaked at 1ppt-2ppts in 2022 and early 2023, but has fallen markedly in the last couple of quarters and fell close to zero in Q1 this year. Like the UK, the indirect effects of energy prices are likely to turn negative in the coming quarters as recent declines in energy prices feed through.

Chart 6: Inflation has fallen especially sharply for items with high energy intensity

Eurozone: Split of core inflation by energy intensity



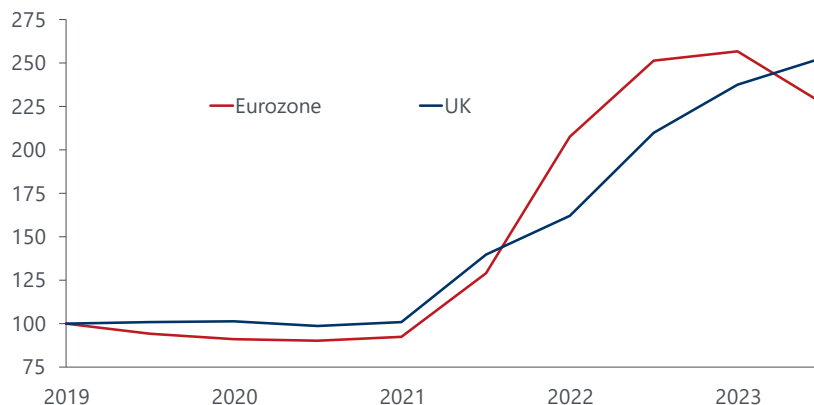
Sources: Oxford Economics/Haver Analytics

Our results for the UK and eurozone are not exactly comparable because of slight differences in the split of items by energy intensity. But the results suggest that indirect effects on core inflation have recently been a bit more persistent in the UK than the eurozone. This is consistent with the more persistent rise in firms' energy costs in the UK, possibly reflecting differences in pricing mechanisms or the share of fixed-price contracts.

Business's energy costs rose faster in the eurozone than the UK between mid-2021 and mid-2023 (with very large increases in some countries such as Italy, Greece, and Estonia) but fell back in H2 last year. In contrast, average energy costs for UK firms in H2 2023 (latest comparable data) were still up from a year earlier and the cumulative rise from H1 2019 to H2 2023 exceeded the eurozone average. Similarly, the direct effects of energy on inflation also came through a bit faster, and faded earlier, in the eurozone compared to the UK.

Chart 7: Businesses' energy costs have been slower to fall in the UK

UK and EZ: Trends in firms' energy costs (weighted 50% gas, 50% electricity for medium users of energy, excluding taxes)
Indexed to 2019 H1 = 100



Sources: Oxford Economics/Haver Analytics

How quickly are indirect effects on core inflation fading?

Over time, trends in firms' energy prices usually have been similar in the UK and eurozone, and followed swings in wholesale costs. The recent divergence between UK and eurozone business energy costs probably reflects lags in UK pricing, rather than a permanent gap. So if wholesale prices are stable (and current levels are well below last year's average), firms' energy costs are likely to fall more in the UK than in the eurozone in the next few quarters, as the UK's recent overshoot unwinds. Hence, while the pass-through from lower wholesale energy prices to consumer prices may be a little quicker in the eurozone in the near term, it may persist a bit longer in the UK.

Implications for central banks

In setting monetary policy, central banks are focussed on persistent trends, rather than temporary swings. The downward indirect effects of energy prices on core inflation will continue for some time, but (unless wholesale energy costs keep falling) probably will fade at around the two-year horizon at which monetary policy usually aims to affect inflation.

Still, this doesn't mean that the indirect effects of energy on core inflation are irrelevant for central banks. First, evidence that some of the recent strength in core and services inflation has reflected temporary effects from energy should help reassure central banks that the more persistent underlying trends in inflation have not been as strong as they might appear at face value. Second, just as the energy-driven surge in inflation in 2022-2023 threatened to destabilise inflation expectations, the unwinding of that surge and past monetary policy tightening should help ensure inflation expectations return to target-consistent levels. This is likely to restrain pay growth and firms' pricing strategies, and thereby (along with sluggish growth) help further dampen the more persistent underlying inflation trends.

We continue to expect the [ECB](#) and [BoE](#) will start to cut interest rates in the next few months, with both cutting several times by year-end and continuing to cut rates next year.